

Experiment 22

Construct a C program to implement the best fit algorithm of memory management.

C Program

```
#include <stdio.h>

int main() {
    int m, n;
    printf("Enter number of memory blocks: ");
    scanf("%d", &m);

    int block[m], process[n];

    printf("Enter block sizes:\n");
    for (int i = 0; i < m; i++)
        scanf("%d", &block[i]);

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter process sizes:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &process[i]);

    printf("\nProcess\tSize\tBlock\n");

    for (int i = 0; i < n; i++) {
        int best = -1;

        for (int j = 0; j < m; j++)
            if (block[j] >= process[i] &&
                (best == -1 || block[j] < block[best]))
                best = j;

        if (best != -1) {
            printf("P%d\t%d\tB%d\n", i + 1, process[i], best + 1);
            block[best] -= process[i];
        } else {
            printf("P%d\t%d\tNot Allocated\n", i + 1, process[i]);
        }
    }

    return 0;
}
```

Sample Output

```
Enter number of memory blocks: 3
Enter block sizes:
100 500 200
Enter number of processes: 3
Enter process sizes:
212 417 112
```

```
Process Size Block
P1      212   B3
P2      417   B2
P3      112  Not Allocated
```

Result

The Best Fit memory allocation algorithm was successfully implemented.